Supporting Information

A Simple, Fast, and Facile Demonstration of a Photochemical Redox Reaction Using Visible Light

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Instruction Notes and Student Guidelines

Table S1: Chemicals used in this demonstration

Chemical Name	CAS Number	Vendor/Material Number	Amount (g)	Price
2,2′-Bipyridine	366-18-7	Millipore Sigma/D216305-10G	10	$23.72
Iron(III) Chloride	7705-08-0	Millipore Sigma/ 157740-5G	5	$13.26
Sodium Pyruvate	113-24-6	Millipore Sigma/ P2256-25G	25	$14.11

Suggested Notes for Instructors

Instructors should verify the experiments before conducting a demonstration in the classrooms. These experiments can be used to demonstrate photochemical reactions as well as oxidation-reduction reactions. Therefore, these experiments can be performed in high schools or universities where we want to reinforce the concepts of photochemistry and oxidation-reduction. These experiments may be relevant if we are teaching photosynthesis and want to show students that light can induce reactions. Crouch et al.[1] have shown that asking students questions to predict the outcome of a demonstration significantly improves the understanding of underlying concepts compared to the students who passively observe the demonstration. Therefore, involving students in the demonstration is important.

An instructor can divide students into a group of 4–6 students per group and let each student do their own experiment. If students want to observe the color change at different times, it is best to prepare several reaction mixtures in the dark and take one reaction at a time and expose to light. For example, if we want to observe color change at 0, 5, 10, 15, and 20 minutes, it is suggested to prepare five reaction mixtures and keep all of them in the dark. Students can then start their stopwatch and keep the reaction mixture in light at 0, 5, 10, 15, and 20 minutes and take a picture at the end of 20 minutes. The reaction mixture that was kept in light at 0 minutes will have an exposure time of 20 minutes, and the reaction mixture that was exposed to light after 20 minutes will have an exposure time of 0 minutes. A side-by-side comparison of the color of the reaction mixtures exposed to light for different times is therefore possible. In addition, students can also perform several control reactions, such as those in **Figure 4**. Instructors may design additional experiments to see if the reaction mixture color intensity changes with light intensity by keeping reaction mixtures at different distances from an LED light source.

While conducting the reactions in this manuscript, we used micropipettes to measure the volume of the reaction mixtures. However, micropipettes are expensive. Instead of using micropipettes, disposable plastic transfer pipettes can be used if the concentration of the stock solutions are adjusted accordingly. We noted that slight changes in reactant concentration do not change the outcomes. We suggest making 0.3 mg/mL 2,2′-bipyridine, 0.1 mg/mL iron(III) chloride, and 0.3 mg/mL sodium pyruvate stock solutions. The reactions can be started by mixing an equal volume of these three solutions, and the total volume of the reaction mixture does not require to be 3 mL.

The followings are some of the recommended questions to ask the students:

1. Typically, in experiments, we keep everything constant and change one variable at a time. Why is this important?
2. What is the difference between UV and visible light? What are the advantages of using visible light as compared to UV light in this experiment?
3. Write the overall reaction.
4. Write the oxidation state of each atom in the reactants and products.
5. In this reaction, which species is oxidized and which species is reduced? Which is the oxidizing agent, and which one is the reducing agent?
6. Why does the color intensity of the reaction mixture increase with time?
7. How can you vary the intensity of light? Do you believe that the intensity of light will influence the rate of reaction? Can you design experiments to show how the intensity of light affects the rate of reaction?
8. What is expected if the vials were kept at different distances from the light source?
9. How would you prove that each reactant is required for the reaction?
10. How would you prove that light is required for this reaction?
11. How would you prove that 2,2′-bipyridine is not required for the reaction but is required for color development?

Student Guidelines

Practice the following before and during the experiments:

1. Read the instruction from your instructor.
2. Know the properties of UV and visible light.
3. Practice writing oxidation states of elements in different compounds.
4. Read the safety data sheets for 2,2′-bipyridine, iron(III) chloride, and sodium pyruvate before using them.
5. Use gloves, lab coat, and safety goggle during the experiments.
6. After the experiment, collect the chemical waste in the hazardous waste containers before disposing of them.

Refrence

(1) Crouch, C.; Fagen, A. P.; Callan, J. P.; Mazur, E. Classroom demonstrations: learning tools or entertainment? *Am. J. Phys.* **2004**, *72* (6), 835-838.